

Why?

***PCA** can't preserve “**local structure**” to reduce dimensions*

How PCA fails?

How t-SNE Works?

✓ Step 1: Compute Pairwise Distances (High Dimension)

- For every point, calculate distance to every other point
- Usually Euclidean distance

✓ Step 2: Convert Distances → Probabilities

- 👉 Nearby points → high probability
- 👉 Far points → low probability

Using Gaussian distribution:

$$P_{ij} = \frac{\exp(-||x_i - x_j||^2)}{\sum \exp(-||x_i - x_k||^2)}$$

✓ Step 3: Create Low-Dimensional Map (2D / 3D)

- Randomly place points

✓ Step 4: Match Similarity in Low Dimension

Uses Student t-distribution:

$$Q_{ij} = \frac{1}{1 + ||y_i - y_j||^2}$$

✓ Step 5: Minimize Difference

👉 Optimize:

$$KL(P||Q)$$

👉 Meaning:

“Make low-dimension distances behave like high-dimension distances”

🎯 Key Insight

t-SNE does NOT preserve:

- exact distances ✗
- global structure ✗

👉 It preserves:

local neighborhood structure